

Magnetic Fields — Paper A

MEDIUM • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Write in sentences — a list of single words scores badly on this topic.
- Every field line you draw must carry an **arrowhead**. A line without one does not score.
- Where a question asks for a number, that number must carry a **unit**.
- The mark for each question is shown in square brackets on the right.
- Reminder: outside a magnet the field lines run from **north to south**, like poles repel and unlike poles attract.

SECTION 1 – MATERIALS AND POLES

1. Name **two** magnetic materials, and **two** metals that are **not** magnetic. [2]

2. Complete the table. Write *yes* or *no* in each empty box. [3]

Material	Attracted by a magnet?
iron	
copper	
nickel	
aluminium	
cobalt	
brass	

3. State what happens in each case. [1]

(a) The north poles of two magnets are brought together.

(b) The south poles of two magnets are brought together. [1]

(c) The north pole of one magnet is brought near the south pole of another. [1]

4. Explain the difference between a **magnetic material** and a **magnet**. [2]

SECTION 2 – THE MAGNETIC FIELD

5. A bar magnet lies on the page with its north pole on the left.

(a) Draw the magnetic field pattern around it. Draw at least four lines. [3]

(b) State which pole your arrows point away from. [1]

6. State **three** rules that must be obeyed when drawing magnetic field lines. [3]

7. On a correct diagram the field lines are drawn closer together at the poles than at the sides. State what this tells you, and why the spacing is worth marks. [2]

8. Describe how you would use a **plotting compass** to plot one magnetic field line around a bar magnet. [3]

SECTION 3 – INDUCED MAGNETISM AND THE EARTH

9. Explain what is meant by **induced magnetism**. [2]

10. State one difference between **soft iron** and **steel**, and name one thing each is used to make. [2]

11. Explain why the needle of a compass always comes to rest pointing north. [2]

12. State the test that proves an object is a magnet, and explain why attraction on its own is not enough. [2]

End of paper. Before you hand it in, go back over every diagram and check that each field line carries an arrowhead pointing from north to south.